RAJALAKSHMI ENGINEERING COLLEGE

Rakchena C 250701576-2025-CSE

R2

Started on	Wednesday, 16 September 2026, 3:52 PM
State	Finished
Completed on	Wednesday, 16 September 2026, 3:56 PM
Time taken	4 mins 1 sec
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int find_floor(int arr[], int low, int high,int x){
    int ans=-1;
    while(low<=high){
        int mid=low+(high-low)/2;
        if(arr[mid]==x){
            return arr[mid];
        }
        if(arr[mid]<x){
            ans= arr[mid];
            low = mid + 1;
        } else {
            high = mid - 1;
        }
    }
    return ans;
}
int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    int arr[n];
    for (int i = 0; i < n; i++) {
        if (scanf("%d", &arr[i]) != 1) return 0;
    }
    int x;
    if (scanf("%d", &x) != 1) return 0;
    int floor_val = find_floor(arr, 0, n - 1, x);
    printf("%d\n", floor_val);
    return 0;
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔

	Input	Expected	Got	
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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